Juyoung Moon

Senior Engineering Manager

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Interactive Online Resume Permanent Residency, Germany (No sponsorship required)

PROFESSIONAL SUMMARY

Engineering leader with 19+ years spanning cloud, platform, data, and quality engineering, from founding a startup to leading enterprise teams in Korea and Germany. At IONOS SE, led three engineering teams (20+ engineers) across IaaS, PaaS, and DBaaS, standardizing QA, delivery, and operational readiness company-wide, including an automation framework that cut test development time by 60%. Currently Global Data Engineering Manager at Samy Alliance, leading a medallion-architecture data platform (dbt, Airflow, BigQuery, Looker) where pipeline optimization improved execution speed by 10% and cut BigQuery infrastructure costs by 42%, while driving agentic AI adoption across analysis and reporting workflows. Started out leading at scale at Samsung, directing a 15-member team that reduced manual validation cycles by 90%. Permanent resident of Germany with a track record of building high-performing, internationally distributed teams.

PROFESSIONAL EXPERIENCE

Samy Alliance

Jun 2026 - Present
Berlin, Germany

Global Data Engineering Manager

- Lead a 5-person team spanning Data Engineering and Data Analysis, setting team roadmap, priorities, and delivery standards for a medallion (bronze/silver/gold) data platform built on BigQuery and Looker, serving analytics, ML, and BI teams for a social media marketing analytics platform
- Drive AI adoption within the team, designing and building agentic AI workflows that consume internal data-pipeline outputs to automate analysis, reporting, and decision support
- Establish internal engineering processes and documentation standards for pipeline development, code review, and data quality, bringing rigor from prior QA leadership into a fast-moving analytics environment
- Own team growth through hiring, onboarding, 1:1s, and performance conversations, coaching engineers and analysts toward greater ownership and technical depth
- Optimized data pipelines end to end, improving execution speed by 10% and reducing BigQuery infrastructure costs by 42% through partitioning, clustering, and caching
- Partner with analytics, ML, and data science leadership to translate business requirements into a prioritized data platform roadmap, defining and launching multiple data products from concept to production
- Serve as the single point of contact for cross-functional stakeholders, aligning engineering delivery with business impact and leading transparent, metrics-based progress communication

IONOS SE

Aug 2018 - Jan 2026
Berlin, Germany

Senior Engineering Manager

Promoted

Jun 2022 - Jan 2026

PEOPLE & ORGANIZATION

- Led three engineering teams totaling 20+ engineers, owning internal platforms and services supporting IaaS, PaaS, and DBaaS products, with full responsibility for delivery, runtime stability, and operational readiness
- Built high-performing engineering team through strategic hiring, onboarding, and mentorship programs
- Conducted structured 1:1s and career development sessions, and implemented a performance review process with SMART goals to support growth and succession planning
- Defined quarterly OKRs and individual KPIs for each team, and led weekly alignment meetings and quarterly reviews to drive accountability, identify blockers, and promote outcome-oriented execution across teams

PLATFORM & DELIVERY

- Integrated Apache Kafka into internal service pipelines to enable event-driven processing, real-time monitoring, and faster incident detection
- Defined roadmap, planning, and priorities for IaaS and PaaS products owned by my teams, incl. snapshot backup, autoscaling, S3, and load balancers
- Worked cross-functionally with product teams on product development, using quality and performance metrics to prioritize improvements as the stakeholder point of contact

QUALITY & AUTOMATION

- Designed and implemented QANG internal Python test automation framework adopted by several teams, reducing test development time by 60% and standardizing testing practices company-wide
- Collaborated with product and infrastructure leadership to define quality KPIs such as mean time to detection (MTTD), mean time to resolution (MTTR), and automated test coverage thresholds
- Established QA metrics dashboard using Grafana and Python scripts, enabling real-time KPI tracking for defect trends, test coverage, and release readiness across multiple teams
- Implemented a reusable performance testing framework to detect system bottlenecks, measuring key metrics like response time, throughput, and I/O latency to improve overall performance by 10%
- Collaborated with cybersecurity teams to ensure ISO 27001 compliance for cloud infrastructure testing processes, incorporating security test automation, CVE scanning, and continuous audit support for internal controls

QA Engineer

Aug 2018 - May 2022

- Led comprehensive testing strategy for cloud infrastructure products including VM, S3, Kubernetes, Load Balancer, and Database services
- Developed modular Python and PyTest test framework increasing test reusability by 70% and reducing CI/CD execution time by 45%
- Managed complex test environments and on-premises hardware provisioning for staging environments supporting 500+ daily test executions
- Designed automated test suites using JMeter, TestCafe, Jenkins, and PostgreSQL achieving 85% test coverage
- Mentored 5+ junior QA engineers through code reviews and best practices training
- Contributed to defining and monitoring KPIs such as flaky test rate, post-release defect rate, and CI reliability, laying the foundation for later leadership responsibilities

Intel Corporation

May 2017 - May 2018
Seoul, South Korea

Customer Quality Engineer

- Served as primary quality contact for major APAC clients managing quality cases for WLAN and CPU product lines including Skylake, Coffee Lake, and Cannon Lake architectures
- Delivered technical support to enterprise customers ensuring 99% issue resolution within SLA timeframes
- Developed diagnostic automation tools using Python, C#, Django, and JavaScript reducing investigation time by 50%
- Managed project budgets up to $50,000 monthly with technical feasibility assessments and ROI recommendations

SanDisk Corporation

Aug 2015 - Sep 2016
Seoul, South Korea

Senior Software Engineer

- Led firmware development team for SATA-based SSD Host Layer translating complex OEM requirements into high-performance embedded solutions
- Designed data transmission layer between host and Flash Translation Layer on ThreadX RTOS improving throughput by 10%
- Implemented automated testing strategies using Python and Jenkins ensuring firmware quality and release readiness
- Conducted low-level debugging and I/O performance tuning to identify firmware bottlenecks, delivering measurable improvements in IOPS and throughput
- Aligned firmware QA goals with corporate milestones and introduced key metrics such as release readiness score and regression failure rate to improve launch predictability
- Established feedback loops with validation and firmware teams, enabling early bug triage and faster regression closure cycles

Samsung Electronics

Jan 2010 - Jul 2015
Hwaseong, South Korea

Software Engineer - Team Lead

Promoted 2013

- Led 15-member engineering team developing automation tools for SSD, UFS, and eMMC testing reducing manual validation cycles by 90%
- Architected enterprise-wide test framework supporting OS installation, monitoring, and automated logging for 1500+ test machines
- Designed scenario-based validation tests for 3D NAND flash memory assessing reliability under thermal stress and endurance conditions
- Developed hardware unit test framework using C, FPGA, oscilloscopes, and logic analyzers for early controller validation
- Defined weekly execution KPIs across validation labs, including hardware fault detection rate and automated execution coverage
- Coordinated cross-site test execution with US and Vietnam teams to maintain milestone alignment
- Introduced dashboard-based KPI tracking to align international QA efforts, including pass rate trends, hardware fault detection metrics, and execution coverage across geographies

BCPI (Best Chemicals & Plastics, Inc.)

Jul 2009 - Dec 2009
Carmona City, Cavite, Philippines

Software Engineering Intern

- Built the company website using ASP.NET and PHP on an Apache web server
- Managed Filipino workers at the company's factory
- Maintained and repaired the factory's legacy COBOL-based automation system

Samsung Software Membership

Jul 2007 - Dec 2009
Busan, South Korea

Software Engineer

- Image Search and Movie Collection Editing Management Program (C#, WPF): Developed an application for image search and movie collection editing and management, using local and web search to enhance the user experience
- 3D Desktop Support Application for Linux Environment (C, C++, Qt): Developed a 3D desktop support application for Linux, implementing features and functionalities that enhance the desktop experience in a 3D environment
- USB Development Emulator (USB Virtual Driver) for Windows (C#, C): Developed a USB virtual driver that emulates USB devices, allowing developers to test and develop USB applications in a Windows environment without the need for hardware
- Video Management and Playback Website and Web-based Video Editing Program (Adobe Flex, Adobe Air): Developed a website for video management and playback plus a web-based video editing program that works without encoding, using Adobe Flex and Adobe Air for a rich and interactive user interface

CORE COMPETENCIES

Category	Technologies & Skills
Leadership & Methodologies	People Leadership · Delivery Management · Agile · Scrum Master · Cross-functional Collaboration · KPI/OKR Definition · Performance Reviews · 1:1 Coaching · Organizational Change · Career Growth Strategy
Data Engineering & Architecture	dbt · Apache Airflow · Google BigQuery · Looker · Medallion Architecture (Bronze/Silver/Gold) · Apache Kafka · Pipeline Optimization · Data Governance
AI & Automation	Agentic AI Workflow Architecture · LLM Pipeline Integration · Tool-use / MCP (Model Context Protocol) · Generative AI Applications · Prompt Engineering & Evaluation · Local LLM Setup · Agent Memory Systems (Hindsight) · Autonomous AI Agents (Hermes) · AI Coding Assistants (Claude Code, Cursor) · Enterprise LLM Platforms (Langdock)
Cloud Platforms & DevOps	AWS · Azure · Kubernetes · Docker · Terraform · Ansible · Jenkins · GitLab CI · GitHub Actions · CI/CD
Programming Languages	Python · Go · Java · Kotlin · C/C++ · C# · JavaScript · TypeScript · Node.js · Shell Scripting
Data Stores & Pipelines	PostgreSQL · MySQL · Redis · MongoDB · DynamoDB · Elasticsearch · RabbitMQ · Power BI · OpenSearch · Y42 · Sanity.io
APIs & Integration	REST · SOAP · WebSocket
Monitoring & Observability	Prometheus · Grafana · ELK Stack (Elasticsearch) · Loki · OpenTelemetry · Flower

Category	Technologies & Skills
Test Automation	PyTest · Selenium · Cypress · TestCafe · JMeter · Cucumber · Playwright · QANG Framework
Compliance & Security	ISO 27001 · GDPR · CVE · Security Testing · Penetration Testing · Risk Assessment · Test Data Governance · Shift-Left Testing

EDUCATION

Degree	Institution	Duration	Location
Bachelor of Engineering in Computer Engineering	Tongmyong University	Mar 2005 - Feb 2010	Busan, South Korea

CERTIFICATIONS

Certification	Issuing Organization	Year
Six Sigma Black Belt	Korea Certification Evaluation	2024
ISTQB Certified Tester Foundation Level (CTFL) 4.0	ISTQB	2025
AWS Certified Cloud Practitioner	AWS	2025
AWS Generative AI Applications	AWS	2025
AWS Cloud Technology Consultant	AWS	2024
Google Cybersecurity Professional Certificate	Google	2024
IBM AI Product Manager Certificate	IBM	2025
Generative AI for Project Managers	IBM	2025
Generative AI for Program Managers	IBM	2025
DevOps, Cloud, and Agile Foundations Specialization	IBM	2025
Strategic Leadership and Management Specialization Business Analytics Specialization Business Value and Project Management Specialization Global Challenges in Business Specialization	University of Illinois Urbana-Champaign	2025
Coaching Skills for Managers Specialization	University of California, Davis	2025
Engineering Project Management Specialization	Rice University	2025

ADDITIONAL EXPERIENCE

Entrepreneurship & Side Projects

- Startup Co-Founder: Photo editing and management software for photo studios, plus a subscription service for pet-related products
- Owned product roadmap and engineering KPIs including time-to-market, customer-reported defects, and release stability, implemented early agile rituals, and drove execution of the MVP within a 3-month window

- Android App Release & Side Projects: Published and maintained mobile utility apps and automation tools (apps-release)

Mentoring & Community

- Mentor for Women Developers (2014 - Present): Mentoring women in tech, currently through individual sessions. Previously ran group mentoring for a Korean government institution and for groups of university and high school students
- Women in Tech & Girls in Tech: Forming an in-house female developer organization, organizing forums, and volunteer activities
- Naver IT Expert: IT-related project and career counseling
- Berlin Korean Professionals Meetup (founded Mar 2018): Created and managed a community for Korean professionals living in Berlin
- Developer Seminars: Ran seminars for developers on Proxy Server, Security Server, and Regular Expressions, and helped university juniors with C language and data structures

Volunteering

- Make-A-Wish Volunteer (2010 - 2014): Activities bringing hope to young children with cancer
- Samsung Electronics Volunteer Leader (2010 - 2012): Planned outreach and participation as volunteering leader, visiting nursing homes and orphanages regularly, teaching violin to children, drawing on sneakers for children in Africa, and painting murals
- Computer Teacher (2007 - 2009): Taught computer language classes as a volunteer for orphaned children at a childcare center